Shaurya Sharma

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Education

Bhagwan Parshuram Institute of Technology — New Delhi, India
Bachelor of Technology in Information Technology — Aug. 2022 – Jun. 2026

Mayo International School — New Delhi, India
Class XII, CBSE — 2021

Technical Skills

Languages: Python, C++, JavaScript, SQL

Web Development: React.js, Node.js, Express.js, MongoDB, RESTful APIs

AI & Generative AI: Generative AI, LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, Embeddings, Chatbot Development

Data & ML: Machine Learning, Pandas, NumPy, Scikit-learn, Data Analysis, Data Visualization

Cloud & DevOps: AWS, Docker, Cloudinary , Git, Cloud-Based Deployment

Core CS: Data Structures, Algorithms, Object-Oriented Programming, Database Management System

Projects

Re-medi — 2025

React 18, Node.js, Express.js, MongoDB Atlas, JWT, Stripe, Razorpay, Cloudinary, Grok API, RAG

- Designed and developed a full-stack MERN healthcare platform comprising **3 role-based portals** (Patient, Doctor, Admin) with appointment scheduling, medicine reminders, dashboards, and analytics.
- Built an **AI-powered medical chatbot** using a complete **RAG pipeline** (document chunking, embeddings, FAISS vector search, semantic retrieval, Grok API), enabling context-aware responses from patient records and medical documents.
- Developed **20+ RESTful APIs** secured with JWT authentication and Bcrypt hashing; integrated **Stripe & Razorpay** for end-to-end payment processing supporting **500+ users**.
- Engineered an MVC backend with **4+ controller modules**, MongoDB Atlas, Cloudinary media storage, and role-based authorization for scalable cloud deployment.

SIMPLE-RAG — 2026

C++, REST API, Ollama (LLM), HNSW, KD-Tree

- Developed a **high-performance vector database** from scratch in **C++**, implementing **3 ANN search algorithms** (**HNSW**, **KD-Tree**, Brute-Force) and supporting **3 distance metrics** (Cosine, Euclidean, Manhattan), reducing search complexity from $O(N)$ **to** $O(\log N)$.
- Built an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline by integrating **Llama 3.2** via **Ollama**, generating **768-dimensional embeddings** for semantic document retrieval and AI-powered question answering with **sub-second search latency**.
- Designed and deployed a **RESTful API** with **12+ endpoints** for document ingestion, vector indexing, similarity search, benchmarking, and RAG inference, alongside a live dashboard visualizing **20+ semantic vectors** using **PCA** for real-time algorithm comparison.

Certifications

- Data Structures & Algorithms – Coding Blocks (2025)
- Web Development Bootcamp – Coding Blocks (2025)
- SQL 50 – LeetCode (2025)

Volunteer Experience

Vice President — Jul. 2024 – Jul. 2025
Rotaract Club of BPIT — New Delhi, India

- Led a team of 120+ members across 15+ community projects.
- Mentored 20+ junior members, improving delivery efficiency by 30%.

Community Services Director — Jul. 2024 – Jul. 2025
Rotaract International District 3012 — New Delhi, India

- Coordinated 25+ Rotaract clubs and district-level initiatives reaching 5,000+ community members.
- Distributed 25,000+ sanitary pads and organized awareness programs across 10+ schools and community centers.